RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 17th June 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS251TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	In Vroom's Expectancy Theory, what does the term Valence refer to?	01	1	3
2.	In Blake and Mouton's Managerial Grid, which leadership style balances concern for people and production?	01	1	3
3.	Mention the contemporary concerns affecting leadership.	02	1	3
4.	What is the role of a Central Bank in the following exchange rate? (a) Fixed exchange (b) Floating exchange	02	2	4
5.	What is the Aggregate Supply–Aggregate Demand (AS-AD) Model?	02	2	4
6.	Name any two main components of Gross Domestic Product (GDP).	02	2	4
	Part – B			
1	A team leader in a tech company has a diverse team with varying skill levels. Some need detailed guidance, while others prefer autonomy. Using Hersey and Blanchard's Situational Leadership Model, suggest leadership styles for different team members based on their maturity levels.	10	3	3
2	Distinguish between hygiene factors and motivation factors. How is Herzberg's theory of motivation useful in practical management	10	3	3
3	At the workplace, workers put inputs into the job, such as education, experience, effort, and energy, and expect to get some outcomes, such as salary, reward, promotion, verbal recognition, and interesting and challenging work, each in equal amounts. Explain the importance of Adam's Equity theory in the given scenario.	10	3	3
4	The economy is experiencing high unemployment rates, and the government is considering cutting taxes to boost consumer spending. Explain in detail how the Keynesian model suggests this tax cut would impact aggregate demand and employment.	10	2	4
5	Identify the key components of Gross Domestic Product (GDP). Evaluate the three primary approaches used to measure a country's GDP.	10	2	4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 17th June 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.
2. Use of Normal Distribution Tables permitted.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	A company is choosing between an 3PL and a 4PL. Illustrate the difference in their roles regarding supply chain "Coordination."	2	3	CO3
2.	Demonstrate why a "Vickrey (Second-Price) Auction" encourages suppliers to bid their true valuation of the contract.	2	2	CO3
3.	Define the three macro processes of the Supply Chain IT framework: CRM, ISCM, and SRM.	2	1	CO4
4.	How "Local Optimization" by different stages leads to a lack of supply chain coordination?	2	2	CO4
5.	Define Vendor-Managed Inventory (VMI) and explain where the decision-making authority lies for replenishment.	2	2	CO4
	Part – B			
1	Compare and contrast the structural behaviors, information requirements, and bidding mechanics of Sealed-Bid First-Price Auctions, English Auctions, Dutch Auctions, and Vickrey (Second-Price) Auctions. Under what explicit market conditions should a purchasing manager select direct negotiation over an automated reverse auction?	10	3	CO3
2	Analyze the structural, behavioral, and technological differences between Direct Procurement and Indirect Procurement infrastructures within an enterprise. Focus your analysis on how their corporate impacts, inventory management styles, and electronic procurement (e-procurement) platform requirements differ.	10	4	CO3
3	Analyze the structural architecture of the Supply Chain IT Framework. Deconstruct the three overarching macro processes—Customer Relationship Management (CRM), Internal Supply Chain Management (ISCM), and Supplier Relationship Management (SRM)—and explain why their structural integration is vital to preventing organizational silos.	10	4	CO4
4	Define the "Bullwhip Effect" in a supply chain. Explain how information distortion occurs as orders move up the supply chain from the retailer to the manufacturer, and describe its impact on supply chain performance.	10	2	CO4
5	The Collaborative Planning, Forecasting, and Replenishment (CPFR) model defines a structured process between supply chain partners. Identify and explain the four major collaborative phases that comprise the CPFR framework.	10	2	CO4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 18th June 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the main goal of user-centered design in cognitive systems?	01	L_2	CO1
2.	Which branch of engineering is historically linked to the origin of cognitive ergonomics?	01	L_1	CO1
3.	What term describes a state of readiness to detect faint, unpredictable signals over long periods?	01	L_2	CO2
4.	What is a "latent error" in a complex system?	01	L_1	CO2
5.	What is confirmation bias?	01	L_2	CO3
6.	Define "Situation Awareness" (SA).	01	L_1	CO3
7.	What is "automation complacency"?	01	L_1	CO2
8.	Give an example of a cognitive tool used to offload memory demands during maintenance tasks.	01	L2	CO3
9.	Define selective attention in the context of system control.	01	L_2	CO3
10.	What is "chunking" in cognitive interface design?	01	L_1	CO3
	Part – B			
1.	Define the "Cognitive Environment" in modern socio-technical systems. Analyze how external environmental stressors degrade an operator's internal mental models and cause system failures.	10	L_4	CO4
2.	Analyze the mechanics of Divided Attention and Time-Sharing during multi-task operations. Explain the Multiple Resources Theory (Wickens) and its application in vehicle cockpit design.	10	L_3	CO3
3a.	A touchscreen ATM has 4 options for withdrawal amount. If the average decision time is 280 ms, what is the decision constant?	05	L_4	CO3
b.	A menu item is 250 mm away from the cursor and has a width of 25 mm. If movement time is measured as 460 ms and a = 100, find b.	05	L_4	CO3
4.	What is Hierarchical Task Analysis (HTA)? Draw and explain an HTA diagram for "withdrawing money from an ATM."	10	L_4	CO3
5.	Discuss Mobile HCI. What unique constraints must a designer consider when moving from desktop to mobile screens?	10	L_4	CO4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 19th June 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Human Resource Management & Analytics		**Course Code : IM364TA**

Part – A

Sl. No.	Questions	M	BT	CO
1.	Differentiate between HR Metrics and HR Analytics.	02	L2	CO4
2.	State any two objectives of an HR Scorecard.	02	L1	CO4
3.	Identify any two-recruitment metrics used in HR Analytics.	02	L1	CO4
4.	List any two benefits of HR Dashboards.	02	L2	CO4
5.	Name any two phases of HR Predictive Modelling.	02	L1	CO4

Part – B

Sl. No.	Questions	M	BT	CO
1.	Explain the alignment of HR Analytics with organizational business goals and strategies. Illustrate with a suitable example.	10	L3	CO4
2.	Discuss the HR Analytics Framework and explain how it supports effective HR decision-making.	10	L3	CO4
3.	Discuss the role of HRIS in supporting HR decision-making and its applications in recruitment, training, and performance management.	10	L3	CO4
4.	Design an HR Dashboard for monitoring recruitment effectiveness. Identify suitable metrics and justify their inclusion.	10	L4	CO4
5.	A technology company has experienced an employee attrition rate of 18% during the past year, significantly higher than the industry average. Available HR data indicate declining employee engagement scores, increased overtime hours, and reduced participation in training programs. Using the phases of HR Predictive Modelling, outline an approach to predict future attrition and recommend suitable retention strategies.	10	L4	CO4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 12.01.2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS251TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	What is a need in the motivation process?	2	2	3
2	State Maslow's hierarchy of needs.	2	1	3
3	Mention any two characteristics of leadership.	2	1	3
4	What is time management?	2	2	2
5	Mention any two stress management techniques.	2	1	2
	Part – B			
1	During an economic slowdown, the central bank reduced interest rates and injected liquidity into the banking system to revive economic activity. a) Explain the role of money and banks in economic recovery. b) Describe how commercial banks influence the money supply.	10	3	4
2	What are the different classifications of unemployment? With a neat diagram explain how it is related to GDP?	10	3	4
3	Explain Keynesian model of Macroeconomics and how it is different from classic model in macroeconomics.	10	2	4
4	Explain Abraham's Maslow's theory of human motivation with hierarchy diagram.	10	2	3
5	What is transactional and transformational leadership? Explain the futures of each type of leadership.	10	3	3

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 23rd June 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management & Economics		**Course Code : HS261TA**

Note: Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Mention the five hierarchical needs proposed by Maslow.	1	1	3
2.	State the two factors that form the basis of leadership styles in the Managerial Grid theory.	1	1	3
3.	Name the two factors in Herzberg's Two-Factor Theory.	1	1	3
4.	An employee feels demotivated because they believe their efforts will not lead to good performance. Which motivation theory explains this situation?	1	3	3
5.	An employee compares their effort-reward ratio with a colleague and feels under-rewarded. Which motivation theory does this refer to?	1	3	3
6.	A team leader adjusts their leadership style based on the maturity level of team members. Which leadership theory is being applied?	1	3	3
7.	Identify the price index used to monitor inflation trends.	1	1	4
8.	Name any two main components of Gross Domestic Product (GDP).	1	1	4
9.	Which macroeconomic model illustrates the relationship between interest rates and output in goods and money markets?	1	2	4
10.	What are the three methods used to calculate GDP?	1	1	4
	Part – B			
1.	Apply Adams' Equity Theory to explain why an employee may feel demotivated despite consistent high performance and limited rewards. What steps can management take to address this perceived inequity?	10	3	3
2.	Explain Hersey and Blanchard's Situational Leadership Theory. How can leaders effectively apply this model to adapt their style based on team members' maturity and competence levels?	10	2	3
3.	A manager follows an autocratic approach, assuming that employees inherently dislike work and require constant supervision, leading to low morale. Identify the theory being applied and evaluate how shifting to an alternative perspective could enhance employee motivation.	10	3	3
4.	Explain the structure of the IS-LM model and discuss its role in analysing equilibrium in the goods and money markets within macroeconomics.	10	2	4
5.	Identify the key components of Gross Domestic Product (GDP). Evaluate the three primary approaches used to measure a country's GDP.	10	2	4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 23rd June 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	The most important operational decision related to transportation in a supply chain is the _____ of deliveries.	1	L2	CO2
2.	Ignoring inventory costs when making _____ decisions can result in choices that worsen the performance of a supply chain.	1	L3	CO2
3.	List the two fundamental supply chain decisions involving transportation and inventory cost trade-off.	1	L2	CO1
4.	How do third parties increase the supply chain surplus? List any two mechanisms.	1	L2	CO3
5.	"Firms must evaluate the risks when they move any function to a third party". Identify two risks for an automobile manufacturing company.	2	L2	CO3
6.	List the processes under CRM.	2	L2	CO4
7.	Illustrate the relationship between the three macro processes and the transaction management foundation.	2	L2	CO4
	Part – B			
1	With the help of a suitable case example elucidate tailored transportation decision made based on size of customer.	10	3	CO2
2	Discuss some auction mechanisms that are often used in practice and highlight some of their feature.	10	3	CO3
3	Supplier Relationship Management (SRM) includes those processes focused on the interaction between the enterprise and suppliers that are upstream in the supply chain. Evaluate.	10	3	CO3
4	Summarize the following managerial actions that increase total supply chain profits and moderate the bullwhip effect.	10	2	CO4
5	Explain the significance of continuous replenishment programs and vendor-managed inventories.	10	4	CO4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 24th June 2025	**CIE – III**	**Max. Marks : 10 + 50**
Semester : 6th	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define static anthropometric data.	01	L_2	CO1
2.	State one principle of workspace design.	01	L_1	CO1
3.	What is meant by 'reach envelope'?	01	L_2	CO2
4.	What does the NIOSH lifting equation help assess?	01	L_1	CO2
5.	State one factor that contributes to low back problems in manual material handling.	01	L_2	CO3
6.	Name the three stages of the information processing model.	01	L_1	CO3
7.	Name two interaction styles in Human-Computer Interaction.	01	L_1	CO2
8.	What does 'GOMS' stand for in interface design?	01	L_1	CO3
9.	In HCI, what is meant by 'affordance'?	01	L_2	CO3
10.	What is meant by 'user-task matching' in HCI?	01	L_1	CO3
	Part – B			
1.	Explain the statistical concepts of percentile and variability in human measurements and their relevance in workspace design.	10	L_3	CO4
2.	Using anthropometric data, design a seated workstation for computer use. Include illustrations of critical dimensions.	10	L_3	CO3
3.	Discuss the characteristics, limitations, and role of working memory in cognitive tasks.	10	L_4	CO4
4.	Discuss the concept of time sharing and divided attention. How can multitasking affect task performance in safety-critical environments?	10	L_4	CO4
5.	Describe the importance of matching interaction styles to users and tasks. Provide examples from mobile applications or web systems.	10	L_4	CO3

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 24th June 2025	**CIE – III**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Human Resource Management & Analytics - IM364TA		

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is an HR Scorecard?	02	L1	CO4
2.	What is the role of regression analysis in HR analytics?	02	L2	CO4
3.	Name any two phases in the HR predictive modelling process.	02	L2	CO4
4.	Name any two software tools used for HR predictive analytics.	02	L1	CO4
5.	What is sentiment analysis and how is it useful in HR analytics?	02	L2	CO4
	Part – B			
1.	Explain how an HR scorecard is developed in an organization. Design a scorecard for talent management function.	10	L2	CO4
2.	You want to design an HR dashboard on performance management issues for the sales force of your organization. Based on your identified list of issues, choose at least two important issues and design an illustrative dashboard.	10	L3	CO4
3.	As an HR manager, you're tasked with designing a data collection process for a training needs analysis. What data sources and methods will you use, and how will you ensure data accuracy?	10	L3	CO4
4.	You are presenting a monthly HR report to the executive team. Choose three key metrics to include and explain how each one contributes to strategic business decisions.	10	L2	CO4
5.	What are the key tools and techniques used in predictive HR analytics? Illustrate with examples how decision trees or logistic regression can solve HR problems.	10	L2	CO4

___*****___

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Go, change the world®

USN

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**HS251TA**	**Duration**	**120 Min**
Sem	**V**	**Improvement CIE**	
PRINCIPLES OF MANAGEMENT AND ECONOMICS			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Which theory compares an individual's input-output ratio to others?	01	1	3
2	In Vroom's Expectancy Theory, what does the term *Valence* refer to?	01	1	3
3	Mention any one difference between transactional leadership from transformational leadership?	01	1	3
4	In Blake and Mouton's Managerial Grid, which leadership style balances concern for people and production?	01	1	2
5	According to McGregor's Theory X, what type of behavior is assumed of employees?	01	1	3
6	In the IS-LM model, what does the IS curve represent?	01	1	5
7	Which Keynesian model incorporates both the goods and labor markets to determine output and employment?	01	1	5
8	What is the key assumption of the Keynesian Cross model?	01	1	5
9	What role does fiscal policy play in the Keynesian Cross model?	01	1	4
10	What does the aggregate supply (AS) curve in the AS-AD model represent?	01	1	5
	Part – B			
1	A mid-sized company is facing low productivity and high turnover due to employees feeling undervalued and lacking growth opportunities. Using Herzberg's Two-Factor Theory, analyze the hygiene factors and motivators that could improve employee satisfaction.	10	4	3
2	A project manager notices that employees' performance drops when they do not understand how their efforts contribute to organizational goals. Using Vroom's Expectancy Theory, explain how the manager can enhance motivation.	10	3	3
3	A team leader in a tech company has a diverse team with varying skill levels. Some need detailed guidance, while others prefer autonomy. Using Hersey and Blanchard's Situational Leadership Model, suggest leadership styles for different team members based on their maturity levels.	10	3	3
4	Describe the Keynesian Cross Model. How does it explain the equilibrium level of national income in an economy? Discuss the impact of changes in government spending and investment on national income in this model.	10	2	4

Academic year 2024-2025 (ODD Sem)

5	Explain the IS-LM model. Discuss the equilibrium in the goods market (IS curve) and money market (LM curve). How do changes in fiscal and monetary policy affect output and interest rates according to this model?	10	2	5

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max	Quiz	–	01	04	01	04	10	–	–	–	–	–
	Marks	Test	–	–	30	10	10	–	20	20	10	–	–